28. Independent Watchdog Timer (IWDT)

28.1 Overview

The Independent Watchdog Timer (IWDT) consists of a 14-bit down counter that must be serviced periodically to prevent counter underflow. The IWDT provides functionality to reset the MCU or to generate a non-maskable interrupt or an underflow interrupt. Because the timer operates with an independent clock from CPU clocks (CPUCLK0 and CPUCLK1) and System clock (ICLK), it is particularly useful in returning the MCU to a known state as a fail-safe mechanism when the system runs out of control. The IWDT can be triggered automatically by a reset, underflow, refresh error, or a refresh of the count value in the registers.

The IWDT functions differ from those of the WDT in the following respects:

The divided IWDT clock (IWDTCLK) is used as the count source (not affected by PCLKB)

Table 28.1 lists the IWDT specifications.

Table 28.1 IWDT specifications

Parameter	Specifications
Count source*1	IWDT clock (IWDTCLK)
Clock division ratio	Division by 1, 16, 32, 64, 128, or 256
Counter operation	Counting down using a 14-bit down-counter
Conditions for starting the counter	● Auto start mode: Counting automatically starts after a reset or after an underflow or refresh error occurs. ● Register start mode: Counting is started with a refresh by writing to the IWDTRR register. ● Only secure developer can select Auto-start mode or Register-start mode.
Conditions for stopping the counter	● Reset (the down-counter and other registers return to their initial values). ● A counter underflows or a refresh error is generated. ● The debug function of primary CPU is enabled.
Window function	Window start and end positions can be specified (refresh-permitted and refresh-prohibited periods).
Independent watchdog timer reset sources	● Down-counter underflows ● Refreshing outside the refresh-permitted period (refresh error)
Non-maskable interrupt/interrupt sources	● Down-counter underflows ● Refreshing outside the refresh-permitted period (refresh error)
Reading of the counter value	The down-counter value can be read by the IWDTSR register.
Event link function (output)	● Down-counter underflow event output ● Refresh error event output
Output signal (internal signal)	● Reset output ● Interrupt request output ● CPU Sleep mode, CPU Deep Sleep mode, Software Standby mode and Deep Software Standby mode count stop control output.
Trust Zone Filter	Security and Privilege attribution can be set.

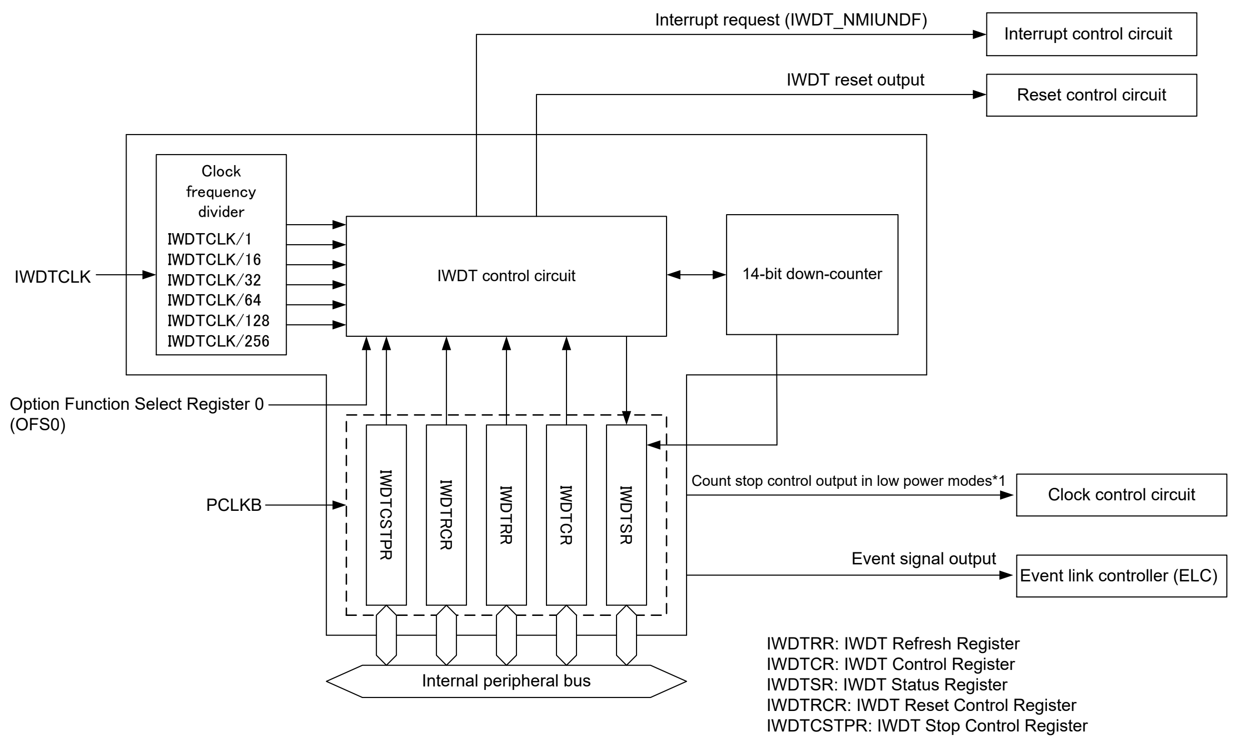
Note 1. Satisfy the frequency of the peripheral module clock (PCLKB) ≧ 32 × (the frequency of the count clock source after division). The bus interface and registers operate with PCLKB, and the 14-bit counter and control circuits operate with IWDTCLK.

28.1.1 Block Diagram

In addition to the peripheral clock (PCLKB), an MCU needs the IWDT clock (IWDTCLK) that does not stop even in low power modes so that the MCU runs even in low power modes in which the peripheral clock (PCLKB) stops. The bus interface and registers operate with the peripheral clock (PCLKB), and the 14-bit down-counter and control circuits operate with the IWDT clock (IWDTCLK).

Signals are connected between the block operating with the peripheral clock (PCLKB) and the block operating with the IWDT clock (IWDTCLK) via a synchronization circuit.

Figure 28.1 shows a block diagram.



Note 1. Low power modes indicate CPU Sleep mode, CPU Deep Sleep mode, Software Standby mode, Deep Software Standby mode 1

Figure 28.1 IWDT block diagram

28.2 Register Specifications

28.2.1 IWDTRR : IWDT Refresh Register

Base address: IWDT = 0x4020_2200
IWDT_NS = 0x5020_2200

Offset address: 0x0

Bit position:	7	6	5	4	3	2	1	0
Bit field:	REFRESH[7:0]							
Value after reset:	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
7:0	REFRESH[7:0]	Refresh Register The counter is refreshed by writing 0x00 and then writing 0xFF to this register.	R/W

Note: S-TYPE-3, P-TYPE-3

The IWDTRR register refreshes the down-counter of the IWDT.

REFRESH[7:0] bits (Refresh Register)

The down-counter of the IWDT is refreshed by writing 0x00 and then writing 0xFF to IWDTRR(refresh operation) within the refresh-permitted period. After the down-counter is refreshed, it starts counting down from the value selected by the IWDT time-out period selection bits (OFS0.IWDTTOPS[1:0]) in the auto start mode.

In the register start mode, counting down starts from the value set by the TOPS[1:0] bits in the IWDT Control Register (IWDTCR).

Also in the register start mode, counting down starts from the value set by the IWDTCR.TOPS[1:0] bits by the first refresh operation after the release from the reset state.

When 0x00 is written, the read value is always 0x00. When a value other than 0x00 is written, the read value is always 0xFF.

For details of the refresh operation, see section 28.3.3. Refresh Operation.

28.2.2 IWDTCR : IWDT Control Register

Base address: IWDT = 0x4020_2200
IWDT_NS = 0x5020_2200

Offset address: 0x2

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	RPSS[1:0]		—	—	RPES[1:0]		CKS[3:0]				—	—	TOPS[1:0]	
Value after reset:	0	0	1	1	0	0	1	1	1	1	1	1	0	0	1	1

Bit	Symbol	Function	R/W
1:0	TOPS[1:0]	Timeout Period Select 0 0: 128 cycles (0x007F) 0 1: 512 cycles (0x01FF) 1 0: 1024 cycles (0x03FF) 1 1: 2048 cycles (0x07FF) Each value in parentheses () denotes a start value of down-counting.	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
7:4	CKS[3:0]	Clock Division Ratio Select 0 0 0 0: IWDTCLK 0 0 1 0: IWDTCLK/16 0 0 1 1: IWDTCLK/32 0 1 0 0: IWDTCLK/64 1 1 1 1: IWDTCLK/128 0 1 0 1: IWDTCLK/256 Other settings are prohibited.	R/W
9:8	RPES[1:0]	Window End Position Select 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (No window end position setting)	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
13:12	RPSS[1:0]	Window Start Position Select 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (No window start position setting)	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The IWDTCR register is used to specify the timeout period (time until the down-counter underflows in the register start mode), clock division ratio, and the window start and end positions of refresh-permitted period. Some restrictions apply to writes to this register. For details, see section 28.3.2. Controlling Writes to the IWDTCR, IWDTRCR, and IWDTCSTPR Registers.

In auto start mode, the settings of this register are disabled and the settings for the option function select register 0 (OFS0) are enabled. The same settings as in the bits in this register can be performed at the option function select register 0 (OFS0). For details, see section 28.3.8. Correspondence Between Option Function Select Register 0 (OFS0) and IWDT Registers.

TOPS[1:0] bits (Timeout Period Select)

The TOPS[1:0] bits select the timeout period, the period until the down-counter underflows, from 128, 512, 1024, and 2048 cycles, taking the divided clock specified in the CKS[3:0] bits as one cycle.

After the down-counter is refreshed, the combination of the CKS[3:0] and TOPS[1:0] bits determines the time (number of IWDT clock (IWDTCLK) cycles) until the counter underflows.

section 28.2.2. IWDTCR : IWDT Control Register shows the relationship among the CKS[3:0] and TOPS[1:0] bits settings, the timeout period, and the number of IWDT clock (IWDTCLK) cycles.

Table 28.2 IWDT timeout period settings

CKS[3:0]				TOPS[1:0]		Clock division ratio	Timeout period (Number of cycles)	Number of IWDT clock (IWDTCLK) cycles
0	0	0	0	0	0	IWDTCLK	128	128
				0	1		512	512
				1	0		1024	1024
				1	1		2048	2048
0	0	1	0	0	0	IWDTCLK/16	128	2048
				0	1		512	8192
				1	0		1024	16384
				1	1		2048	32768
0	0	1	1	0	0	IWDTCLK/32	128	4096
				0	1		512	16384
				1	0		1024	32768
				1	1		2048	65536
0	1	0	0	0	0	IWDTCLK/64	128	8192
				0	1		512	32768
				1	0		1024	65536
				1	1		2048	131072
1	1	1	1	0	0	IWDTCLK/128	128	16384
				0	1		512	65536
				1	0		1024	131072
				1	1		2048	262144
0	1	0	1	0	0	IWDTCLK/256	128	32768
				0	1		512	131072
				1	0		1024	262144
				1	1		2048	524288

CKS[3:0] bits (Clock Division Ratio Select)

These bits select IWDT clock (IWDTCLK) division ratio from among division by 1, 16, 32, 64, 128, and 256.

Combined with the TOPS[1:0] bits setting, a count period between 128 and 524288 cycles of the IWDT clock (IWDTCLK) can be selected for the IWDT.

Note: In order to read the down-counter value correctly, the relationship between the peripheral clock (PCLKB) frequency and the count clock (IWDTCLK) frequency must be set appropriately. For the setting, see Note1 in Table 28.1

RPES[1:0] bits (Window End Position Select)

These bits select 75%, 50%, 25% or 0% of the count period for the window end position of the down-counter. The selected window end position should be a value smaller than the value for the window start position (that is, window start position > window end position). If the value for the window end position is greater than the value for the window start position, only the value for the window start position is effective.

The counter values for the window start and end positions selected by setting the RPES[1:0] and RPSS[1:0] bits change depending on the TOPS[1:0] bits setting.

Table 28.3 lists the counter values for the window start and end positions corresponding to TOPS[1:0] bits values.

Table 28.3 Relationship between timeout periods and window start and end counter values

TOPS[1:0]		Timeout period		Window start and end counter value			
		Number of cycles	Counter value	100%	75%	50%	25%
0	0	128	0x007F	0x007F	0x005F	0x003F	0x001F
0	1	512	0x01FF	0x01FF	0x017F	0x00FF	0x007F
1	0	1024	0x03FF	0x03FF	0x02FF	0x01FF	0x00FF
1	1	2048	0x07FF	0x07FF	0x05FF	0x03FF	0x01FF

RPSS[1:0] bits (Window Start Position Select)

These bits select a counter window start position from 100%, 75%, 50%, or 25% of the count period (100% when the count starts and 0% when the counter underflows). The interval between the window start position and window end position is the refresh-permitted period and the other periods are refresh-prohibited periods.

Figure 28.2 shows the relationship between the RPSS[1:0] and RPES[1:0] bits settings and the refresh-permitted and refresh-prohibited periods.

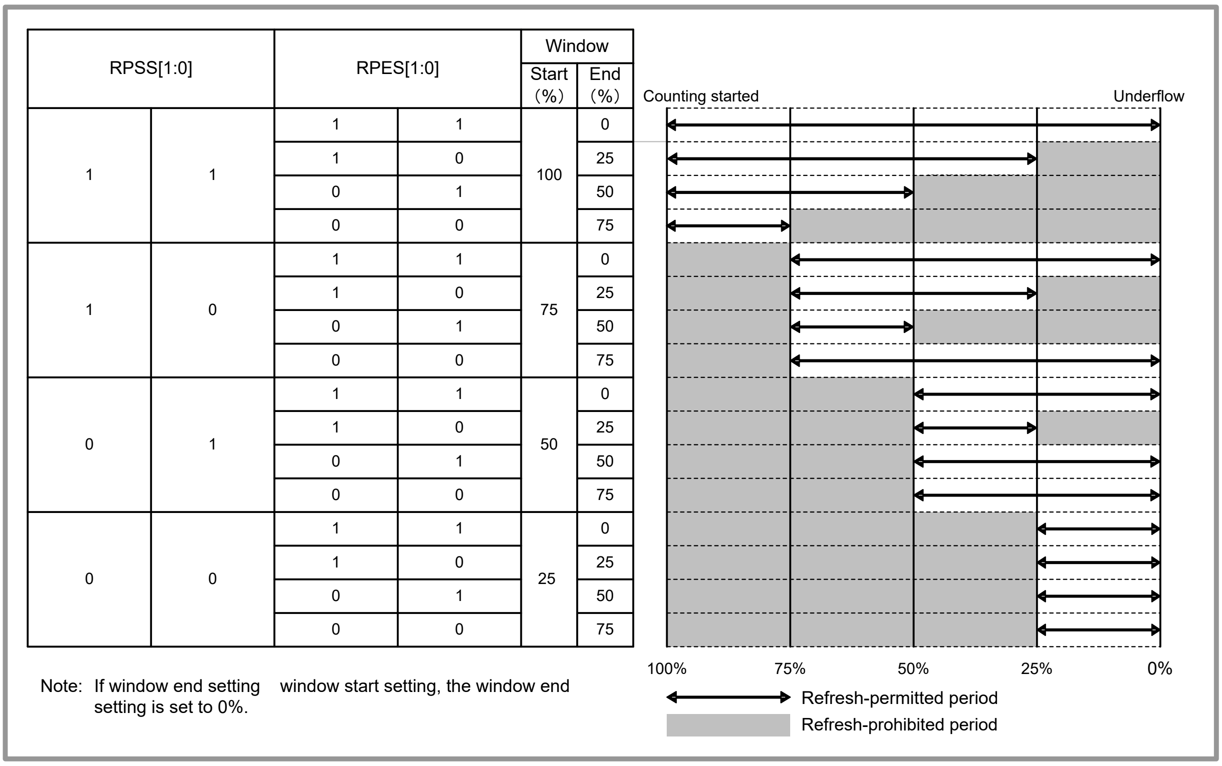


Figure 28.2 RPSS[1:0] and RPES[1:0] bits settings and refresh-permitted and refresh-prohibited periods

28.2.3 IWDTSR : IWDT Status Register

Base address: IWDT = 0x4020_2200
IWDT_NS = 0x5020_2200

Offset address: 0x4

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	REFEF	UNDFF	CNTVAL[13:0]													
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
13:0	CNTVAL[13:0]	Down-Counter Value Counter value of the down-counter	R
14	UNDFF	Underflow Flag 0: No underflow occurred. 1: Underflow occurred.	R/W
15	REFEF	Refresh Error Flag 0: No refresh error occurred. 1: Refresh error occurred.	R/W

Note: S-TYPE-3, P-TYPE-3

The IWDTSR register indicates the counter value of the down-counter and whether an underflow or refresh error occurred in the down-counter.

It is initialized by the reset source of the IWDT. IWDTSR is not initialized by other reset sources.

CNTVAL[13:0] bits (Down-Counter Value)

Read these bits to confirm the counter value of the down-counter.

Note: The read value might differ from the actual count by 1.

UNDFF flag (Underflow Flag)

Read the UNDFF flag to confirm whether an underflow occurred in the down-counter.

A value of 1 indicates that the down-counter underflowed. A value of 0 indicates that the down-counter has not underflowed.

Write 0 to the UNDFF flag to clear the value. Writing 1 has no effect.

Clearing of the UNDFF flag takes (N + 2) IWDTCLK cycles and 32 PCLKB cycles. In addition, clearing of this flag is ignored for (N + 2) IWDTCLK cycles after an underflow. N is specified as follows:

Auto start mode:

- When OFS0.IWDTCKS[3:0] = 0x0, N = 1
- When OFS0.IWDTCKS[3:0] = 0x2, N = 16
- When OFS0.IWDTCKS[3:0] = 0x3, N = 32
- When OFS0.IWDTCKS[3:0] = 0x4, N = 64
- When OFS0.IWDTCKS[3:0] = 0xF, N = 128
- When OFS0.IWDTCKS[3:0] = 0x5, N = 256

Register start mode:

- When IWDTCR.CKS[3:0] = 0x0, N = 1
- When IWDTCR.CKS[3:0] = 0x2, N = 16
- When IWDTCR.CKS[3:0] = 0x3, N = 32
- When IWDTCR.CKS[3:0] = 0x4, N = 64
- When IWDTCR.CKS[3:0] = 0xF, N = 128
- When IWDTCR.CKS[3:0] = 0x5, N = 256

REFEF flag (Refresh Error Flag)

Read the REFEF flag to confirm whether a refresh error (a refresh operation was performed during the prohibited period) occurred.

A value of 1 indicates that a refresh error occurred. A value of 0 indicates that no refresh error has occurred.

Write 0 to the REFEF flag to clear the value. Writing 1 has no effect.

Clearing of the REFEF flag takes (N + 2) IWDTCLK cycles and 32 PCLKB cycles. In addition, clearing of this flag is ignored for (N + 2) IWDTCLK cycles following a refresh error. N is specified as follows:

Auto start mode:

- When OFS0.IWDTCKS[3:0] = 0x0, N = 1
- When OFS0.IWDTCKS[3:0] = 0x2, N = 16
- When OFS0.IWDTCKS[3:0] = 0x3, N = 32
- When OFS0.IWDTCKS[3:0] = 0x4, N = 64
- When OFS0.IWDTCKS[3:0] = 0xF, N = 128
- When OFS0.IWDTCKS[3:0] = 0x5, N = 256

Register start mode:

- When IWDTCR.CKS[3:0] = 0x0, N = 1
- When IWDTCR.CKS[3:0] = 0x2, N = 16
- When IWDTCR.CKS[3:0] = 0x3, N = 32
- When IWDTCR.CKS[3:0] = 0x4, N = 64
- When IWDTCR.CKS[3:0] = 0xF, N = 128
- When IWDTCR.CKS[3:0] = 0x5, N = 256

28.2.4 IWDTRCR : IWDT Reset Control Register

Base address: IWDT = 0x4020_2200
IWDT_NS = 0x5020_2200

Offset address: 0x6

Bit position:	7	6	5	4	3	2	1	0
Bit field:	RSTIRQS	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.	R/W
7	RSTIRQS	Reset Interrupt Request Select 0: Enable non-maskable interrupt requests or interrupt request output. 1: Enable reset output.	R/W

Note: S-TYPE-3, P-TYPE-3

IWDTRCR register controls the reset output or interrupt request output when the down-counter of the IWDT underflows. Some restrictions apply to writes to this register. For details, see section 28.3.2. Controlling Writes to the IWDTCR, IWDTRCR, and IWDTCSTPR Registers.

In auto start mode, the settings of this register are disabled and the settings for the input ports (OFS0) are enabled. The same settings as in the bits in this register can be performed at the input ports (OFS0). For details, see section 28.3.8. Correspondence Between Option Function Select Register 0 (OFS0) and IWDT Registers.

RSTIRQS bit (Reset Interrupt Request Select)

The RSTIRQS bit selects the reset output or interrupt request output when the down-counter underflows or a refresh error occurs.

28.2.5 IWDTCSTPR : IWDT Count Stop Control Register

Base address: IWDT = 0x4020_2200
IWDT_NS = 0x5020_2200

Offset address: 0x8

Bit position:	7	6	5	4	3	2	1	0
Bit field:	SLCSTP	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	—	This bit is read as 0.	R
7	SLCSTP	CPU Sleep-Mode Count Stop Select 0: Disable count stop. 1: Stop the counter when the primary CPU enters CPU Sleep mode or CPU Deep Sleep mode, and when the MCU enters Software Standby mode or Deep Software Standby mode 1.	R/W

Note: S-TYPE-3, P-TYPE-3

The IWDTCSTPR register is used to stop the down-counter of the IWDT in case the primary CPU enters low-power modes. Some restrictions apply to writes to this register. For details, see section 28.3.2. Controlling Writes to the IWDTCR, IWDTRCR, and IWDTCSTPR Registers.

In auto start mode, the settings of this register are disabled and the settings for the option function select register 0 (OFS0) are enabled. The same settings as in the bits in this register can be performed at the option function select register 0 (OFS0). For details, see section 28.3.8. Correspondence Between Option Function Select Register 0 (OFS0) and IWDT Registers.

SLCSTP bit (CPU Sleep-Mode Count Stop Select)

The SLCSTP bit selects whether to stop the counter in case the primary CPU enters CPU Sleep mode or CPU Deep Sleep mode, and the MCU enters Software Standby mode or Deep Software Standby mode 1.

28.3 Operation

28.3.1 Count Operation in Each Start Mode

The start mode of the IWDT is selected by setting the IWDT start mode selection (OFS0.IWDTSTRT) bit during a reset.

When the IWDT start mode selection (OFS0.IWDTSTRT) bit is 1 (register start mode), the IWDT Control Register (IWDTCR), IWDT Reset Control Register (IWDTRCR), and IWDT Count Stop Control Register (IWDTCSTPR) are enabled, and counting is started by writing to the IWDT Refresh Register (IWDTRR) to refresh the counter. When the IWDT start mode selection (OFS0.IWDTSTRT) bit is 0 (auto start mode), the settings of the option function select register 0 (OFS0) are enabled, and counting automatically starts after a reset.

(1) Register start mode

When the register start mode is selected by setting the IWDT start mode selection (OFS0.IWDTSTRT) bit to 1 during a reset, the IWDT Control Register (IWDTCR), IWDT Reset Control Register (IWDTRCR), and IWDT Count Stop Control Register (IWDTCSTPR) are enabled.

After the reset state is released, set the clock division ratio, window start and end positions, and timeout period in the IWDTCR register, the reset output or interrupt request output in the IWDTRCR register, and the IWDT down-counter stop control at transitions to low power modes in the IWDTCSTPR register. Then, refresh the down-counter to set the value specified by the IWDTCR.TOPS[1:0] bits in the counter and start counting down.

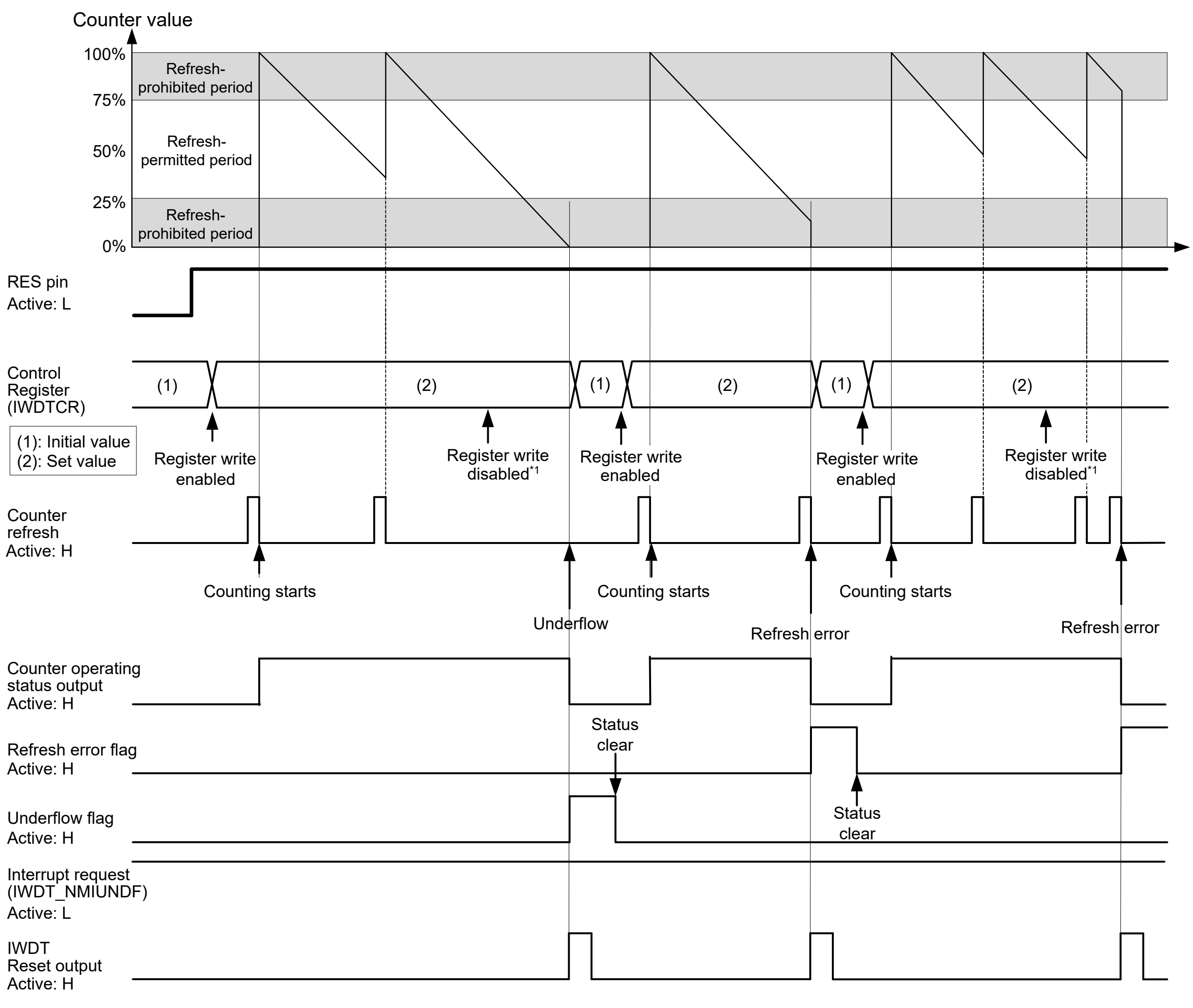
After that, as long as the program continues normal operation and the counter is refreshed in the refresh permitted period, the value in the counter is reset each time the counter is refreshed and counting down continues. The IWDT does not output a reset signal as long as this continues.

However, if the down counter underflows because the down-counter cannot be refreshed due to a program runaway, or if a refresh error occurs because the counter was refreshed outside the refresh-permitted period, the IWDT outputs a reset signal or a non-maskable interrupt request (IWDT_NMIUNDF).

The reset output or interrupt request output is selected by setting the IWDTRCR.RSTIRQS bit.

Figure 28.3 shows an example of operation under the following conditions:

- Register start mode (OFS0.IWDTSTRT = 1)
- Reset output is enabled (IWDTRCR.RSTIRQS = 1)
- The window start position is 75% (IWDTCR.RPSS[1:0] = 10b)
- The window end position is 25% (IWDTCR.RPES[1:0] = 10b)



Note 1. See section 28.3.2. Controlling Writes to the IWDTCR, IWDTRCR, and IWDTCSTPR Registers

Figure 28.3 Operation example in register start mode

(2) Auto start mode

When the auto start mode is selected by setting the IWDT start mode selection (OFS0.IWDTSTRT) bit to 0 during a reset, the IWDT Control Register (IWDTCR), IWDT Reset Control Register (IWDTRCR), and IWDT Count Stop Control Register (IWDTCSTPR) are disabled.

During a reset, the clock division ratio, window start and end positions, timeout period, reset output or interrupt request output, and counter stop control at transitions to low power modes should be set through the option function select register 0

(OFS0). Then, when the reset state is released, the timeout period set through the option function select register 0 (OFS0) is set in the down-counter, and then the down-counter automatically starts counting down.

After that, as long as the program continues normal operation and the counter is refreshed in the refresh permitted period, the value in the counter is reset each time the counter is refreshed and counting down continues. The IWDT does not output a reset signal as long as this continues.

However, if the down-counter underflows because the down counter cannot be refreshed due to a program runaway, or if a refresh error occurs because the counter was refreshed outside the refresh-permitted period, the IWDT outputs a reset signal or a non-maskable interrupt request (IWDT_NMIUNDF).

The reset output or interrupt request output is selected by setting the IWDT reset interrupt request select bit (OFS0.IWDTRSTIRQS).

Figure 28.4 shows an example of operation under the following conditions:

- Auto start mode (OFS0.IWDTSTRT = 0)
- Non-maskable interrupt request output is enabled (OFS0.IWDTRSTIRQS = 0)
- The window start position is 75% (OFS0.IWDTRPSS[1:0] = 10b)
- The window end position is 25% (OFS0.IWDTRPES[1:0] = 10b)

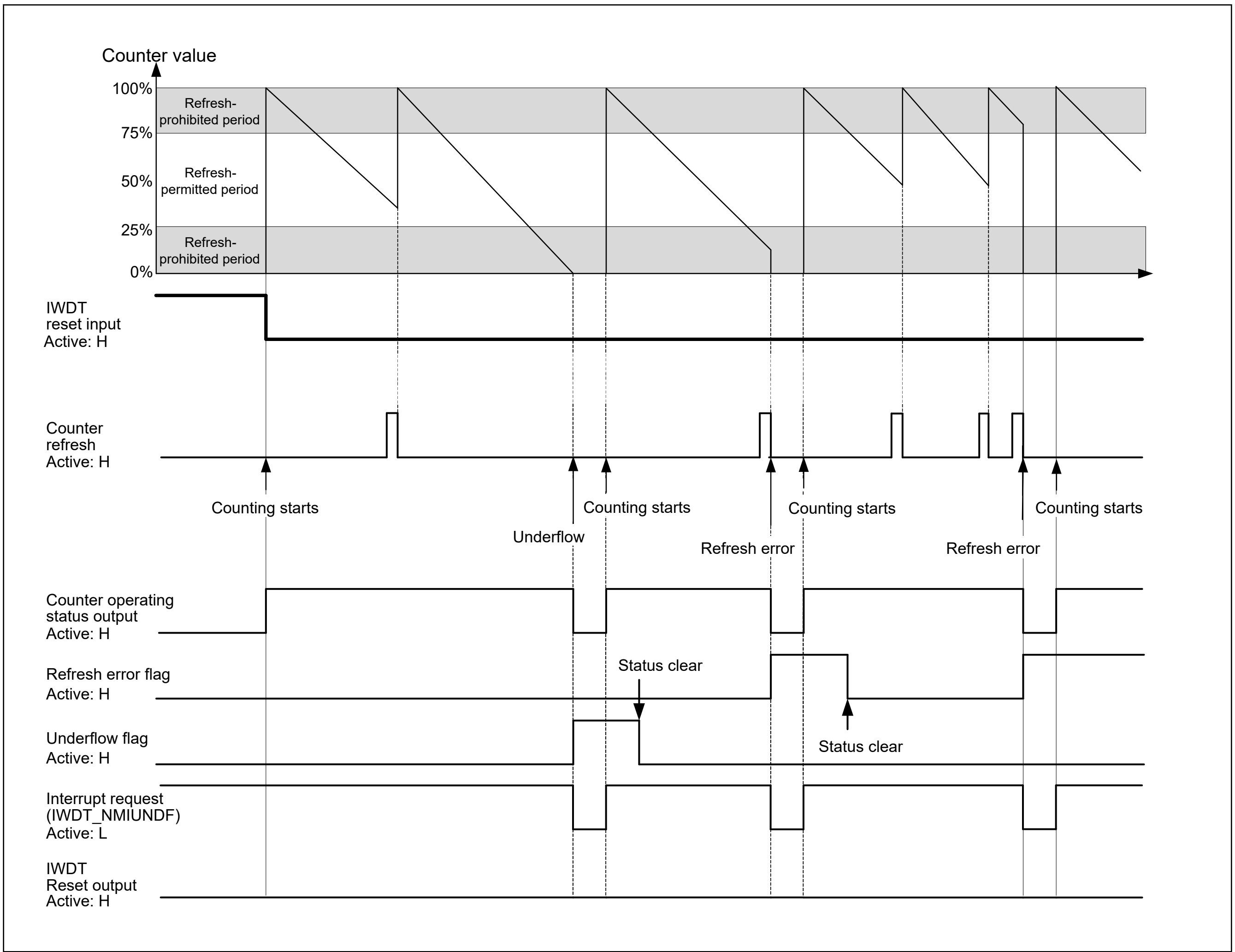


Figure 28.4 Operation example in auto start mode

28.3.2 Controlling Writes to the IWDTCR, IWDTRCR, and IWDTCSTPR Registers

Writing to the IWDT Control Register (IWDTCR) is possible only once between the release from the reset state and the first refresh operation.

After a refresh operation (counting starts) or by writing to the IWDTCR register, the protection signal in the IWDT becomes 1 to protect the IWDTCR register against subsequent attempts of writing.

The IWDT Reset Control Register (IWDTRCR) and the IWDT Count Stop Control Register (IWDTCSTPR) are also controlled in the same way.

This protection is released by a reset source to the IWDT. With other reset sources, the protection is not released.

Figure 28.5 shows control waveforms produced in response to writing to the IWDTCR register.

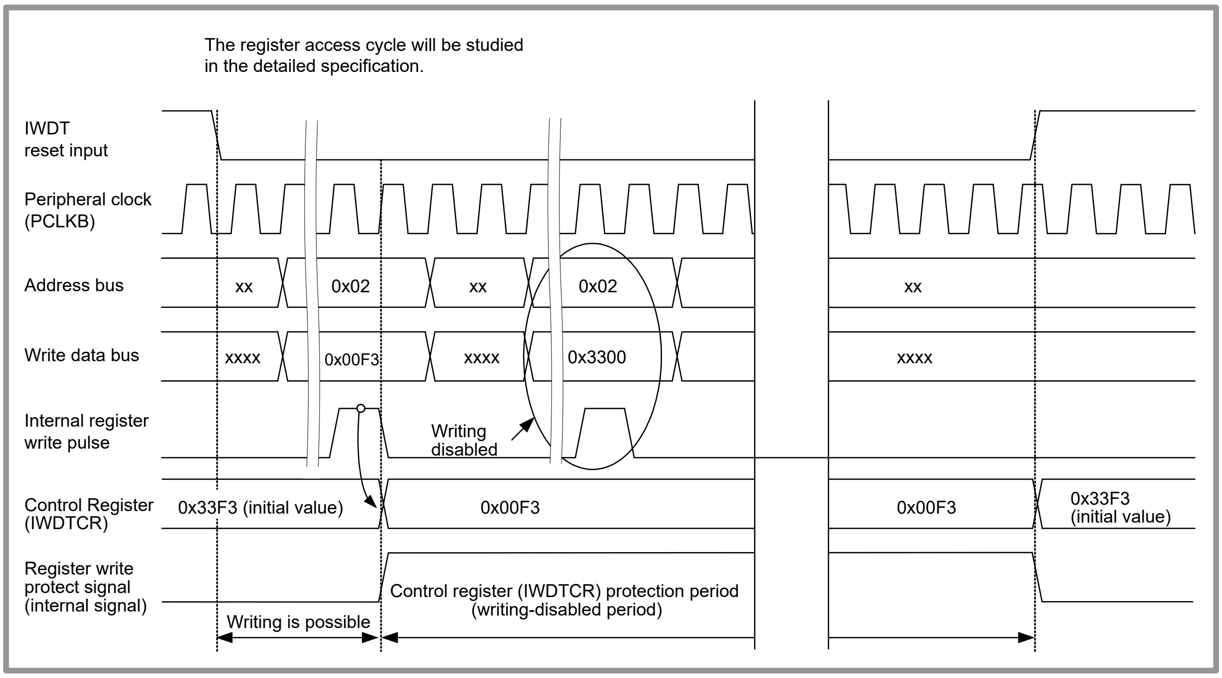


Figure 28.5 Control waveforms produced in response to writes to the IWDTCR register

28.3.3 Refresh Operation

The down-counter is refreshed and starts operation (counting is started by refreshing) by writing the values 0x00 and then 0xFF to the IWDT Refresh Register (IWDTRR). If a value other than 0xFF is written after 0x00, the counter is not refreshed. After such invalid writing, correct refreshing is performed by writing 0x00 and then 0xFF again to the IWDT Refresh Register (IWDTRR).

When writing is done in the order of 0x00 (first time) → 0x00 (second time), and if 0xFF is written after that, the writing order 0x00 → 0xFF is satisfied; writing 0x00 ((n–1) th time) → 0x00 (nth time) → 0xFF is valid and correct refreshing will be done. Even when the first value written before 0x00 is not 0x00, correct refreshing will be done if the operation contains the sequence of writing 0x00 → 0xFF. Correct refreshing is also performed when a register other than IWDTRR is accessed or IWDTRR is read between writing 0x00 and writing 0xFF to IWDTRR.

[Example write sequences that are valid for refreshing the counter]

- 0x00 → 0xFF
- 0x00 ((n-1) th time) → 0x00 (nth time) → 0xFF
- 0x00 → access to another register or read from IWDTRR → 0xFF

[Example write sequences that are invalid for refreshing the counter]

- 0x23 (a value other than 0x00) → 0xFF
- 0x00 → 0x54 (a value other than 0xFF)
- 0x00 → 0xAA (0x00 and a value other than 0xFF) → 0xFF

Even when 0x00 is written to IWDTRR outside the refresh-permitted period, if 0xFF is written to IWDTRR in the refresh-permitted period, the writing sequence is valid and refreshing will be done.

After 0xFF is written to the IWDTRR register, refreshing the counter requires up to four cycles of the signal for counting (the clock division ratio select bits (IWDTCR.CKS[3:0]) determine how many cycles of the IWDT clock (IWDTCLK) make up one cycle for counting).

Therefore, writing 0xFF to the IWDTRR register should be completed four count cycles before the end position of the refresh-permitted period or the down-counter underflow. The value of the down-counter can be checked using the down-counter value bits (IWDTSR.CNTVAL[13:0]).

[Example refreshing timings]

- When the window start position is set to 0x03FF, even if 0x00 is written to IWDTRR before 0x03FF is reached (at 0x0402, for example), refreshing occurs if 0xFF is written to the IWDTRR register after the value of the IWDTSR.CNTVAL[13:0] bits reaches 0x03FF.
- When the window end position is set to 0x03FF, refreshing is done if 0x0403 (four count cycles before 0x03FF) or a greater value is read from the IWDTSR.CNTVAL[13:0] bits immediately after writing 0x00 → 0xFF to the IWDTRR register.
- When the refresh-permitted period continues until count 0x0000, refreshing can be done immediately before an underflow. In this case, if 0x0003 (four count cycles before an underflow) or a greater value is read from the IWDTSR.CNTVAL[13:0] bits immediately after writing 0x00 → 0xFF to the IWDTRR register, no underflow occurs and refreshing is done.

Figure 28.6 shows the IWDT refresh operation waveforms when the peripheral clock (PCLKB) > IWDT clock (IWDTCLK) and the clock division ratio = IWDTCLK/1. Figure 28.7 shows the IWDT refresh operation waveforms when the peripheral clock (PCLKB) < IWDT clock (IWDTCLK) and the clock division ratio = IWDTCLK/16.

When transitioning to Software Standby mode or Deep Software Standby mode 1, the refresh procedure must be completed and the read value of the IWDTRR register must be 0xFF before the transitioning (Write 0xFF after writing 0x00 to the IWDTRR register, and confirm that the read value of the IWDTRR register is 0xFF).

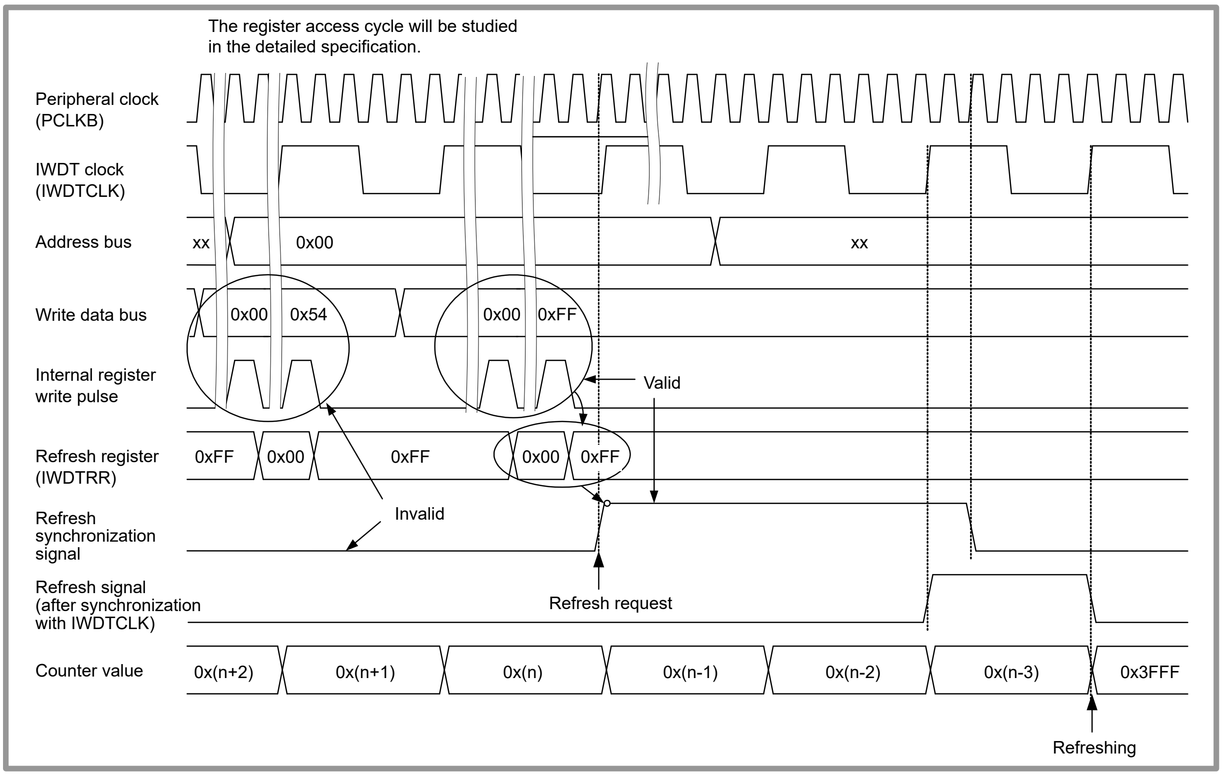


Figure 28.6 IWDT refresh operation waveforms (IWDTCR.CKS[3:0] = 0000b, IWDTCR.TOPS[1:0] = 11b)

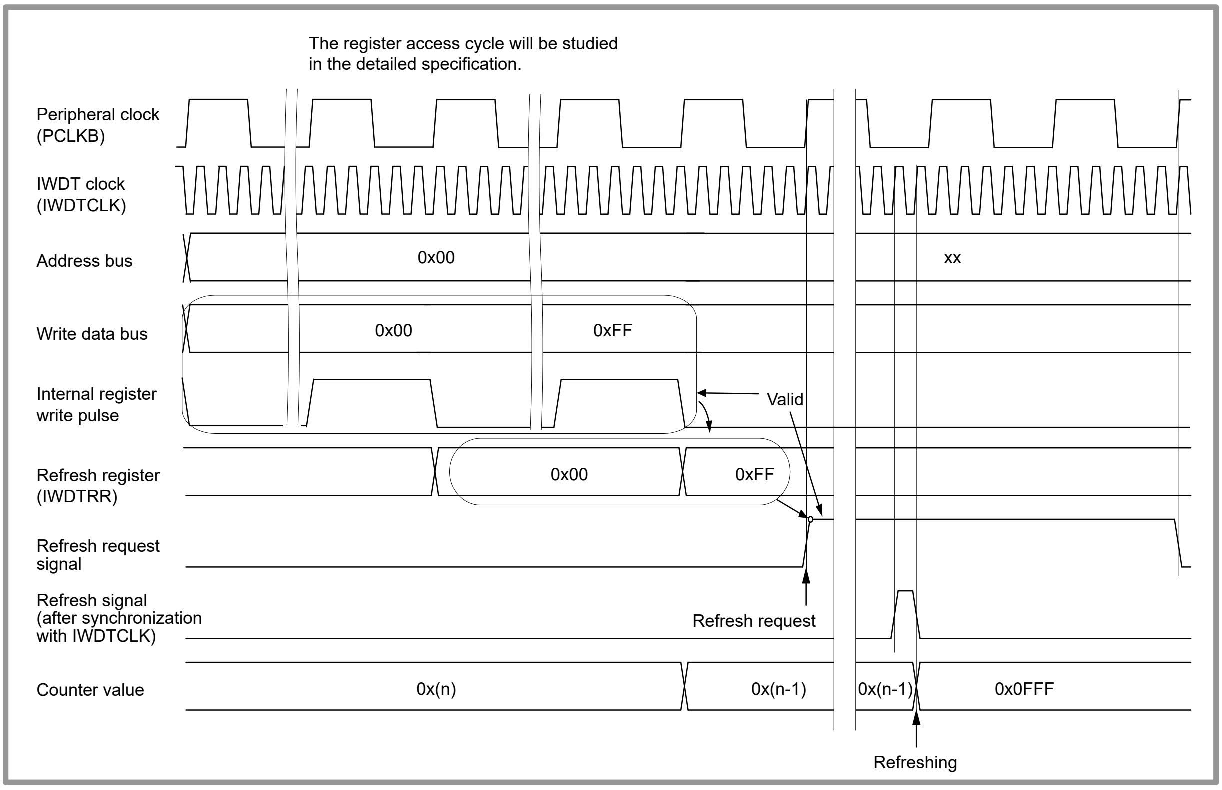


Figure 28.7 IWDT refresh operation waveforms (IWDTCR.CKS[3:0] = 0010b, IWDTCR.TOPS[1:0] = 01b)

28.3.4 Status Flags

The refresh error flag (IWDTSR.REFEF) and underflow flag (IWDTSR.UNDFF) retain the source of the reset output from the IWDT or the source of the interrupt request from the IWDT.

Thus, after release from the reset state or interrupt request generation, read the IWDTSR.REFEF and IWDTSR.UNDFF flags to check for the reset or interrupt source.

For each flag, writing 0 clears the bit and writing 1 has no effect.

Leaving each flag unchanged does not affect operation. If the flags are not cleared, the earlier reset or interrupt source is cleared and the new reset or interrupt source is written when the IWDT outputs a reset signal next.

28.3.5 Reset Output

When the reset interrupt select bit (IWDTRCR.RSTIRQS) is set to 1 in register start mode or when the IWDT reset interrupt request select bit (OFS0.IWDTRSTIRQS) is set to 1 in auto start mode, a reset signal is output for one count cycle when the down counter overflows or a refresh error occurs. In register start mode, the down counter is initialized (all bits set to 0) and kept in that state after assertion of the reset signal. After the reset is released and the program is restarted, the counter is set up again and counting down is started by refreshing. In auto start mode, counting down automatically starts after the reset output.

28.3.6 Interrupt Sources

When the reset interrupt select bit (IWDTRCR.RSTIRQS) is set to 0 in register start mode or when the IWDT reset interrupt request select bit (OFS0.IWDTRSTIRQS) is set to 0 in auto start mode, a non-maskable interrupt (IWDT_NMIUNDF) is generated for one-count cycle when the down-counter overflows or a refresh error occurs.

28.3.7 Reading the Down-counter Value

Because the down-counter in the IWDT operates on the IWDT clock (IWDTCLK), the counter value cannot be read directly.

Therefore, the IWDT synchronizes the counter value with the peripheral clock (PCLKB) and stores it in the down-counter value bits (CNTVAL[13:0]) of the IWDT Status Register (IWDTSR). The counter value can be checked indirectly by reading the value stored in the IWDTSR.CNTVAL[13:0] bits.

Reading the counter value requires multiple peripheral clock (PCLKB) cycles (up to four clock cycles), and the read counter value may differ from the actual counter value by a value of one count.

Figure 28.8 shows the processing for reading the IWDT counter value when the peripheral clock (PCLKB) > IWDT clock (IWDTCLK) and the clock division ratio = IWDTCLK/1. Figure 28.9 shows the processing for reading the IWDT counter value when the peripheral clock (PCLKB) < IWDT clock (IWDTCLK) and the clock division ratio = IWDTCLK/16.

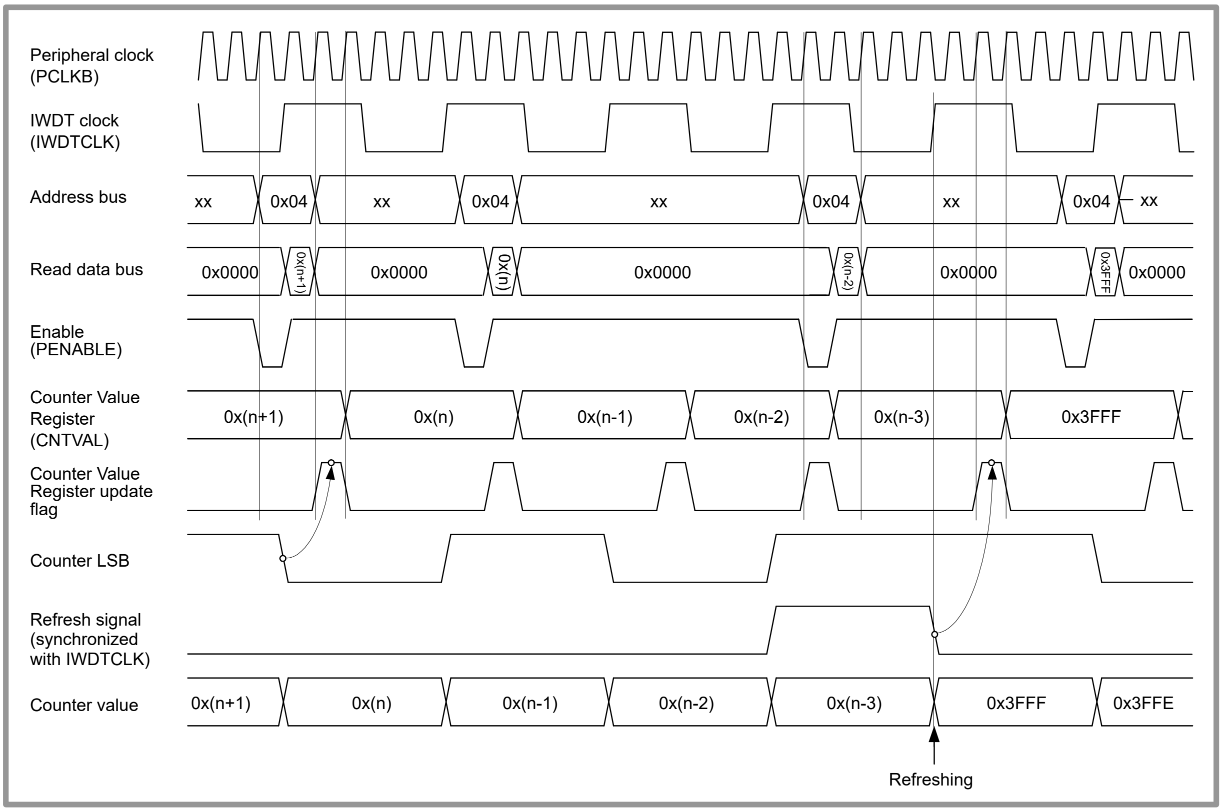


Figure 28.8 Processing for reading IWDT counter value when IWDTCR.CKS[3:0] = 0000b and IWDTCR.TOPS[1:0] = 11b

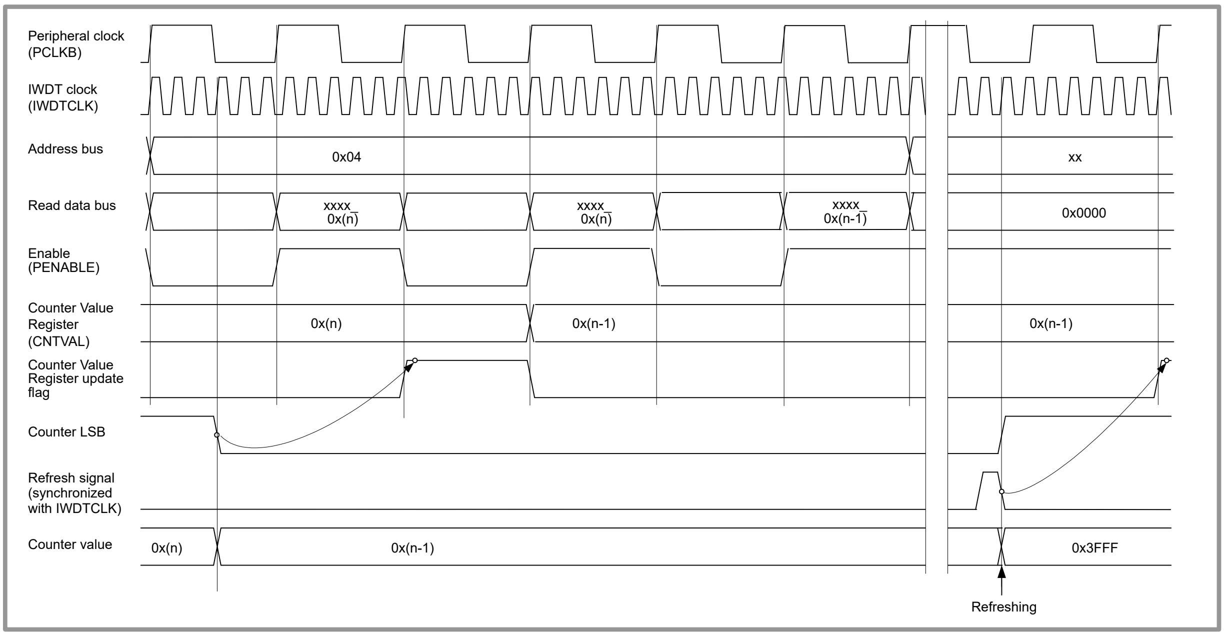


Figure 28.9 Processing for reading IWDT counter value when IWDTCR.CKS[3:0] = 0010b and IWDTCR.TOPS[1:0] = 11b

28.3.8 Correspondence Between Option Function Select Register 0 (OFS0) and IWDT Registers

Table 28.4 lists the correspondence between the option function select register 0 (OFS0) and the IWDT registers (IWDT control register (IWDTCR), IWDT reset control register (IWDTRCR), and IWDT count stop control register (IWDTCSTPR)) regarding control of the down-counter, reset or interrupt request output, and count stop function.

Control can be switched between the option function select register 0 (OFS0) and the IWDT registers (IWDTCR, IWDTRCR, and IWDTCSTPR) through the setting of the IWDT start mode selection (OFS0.IWDTSTRT) bit.

Note that the option function select register 0 (OFS0) setting should be kept unchanged during IWDT operation.

Table 28.4 Association between Option Function Select Register 0 (OFS0) and the IWDT register

Control	Function	OFS0 register (enabled in auto start mode) OFS0.IWDTSTRT = 0	IWDT register (enabled in register start mode) OFS0.IWDTSTRT = 1
Down-counter	Timeout period selection	OFS0.IWDTTOPS[1:0]	IWDTCR.TOPS[1:0]
	Clock division ratio selection	OFS0.IWDTCKS[3:0]	IWDTCR.CKS[3:0]
	Window start position selection	OFS0.IWDTRPSS[1:0]	IWDTCR.RPSS[1:0]
	Window end position selection	OFS0.IWDTRPES[1:0]	IWDTCR.RPES[1:0]
Reset output/ interrupt request output	Reset output/ interrupt request output select	OFS0.IWDTRSTIRQS	IWDTRCR.RSTIRQS
Stop the counter	CPU Sleep-mode count stop selection	OFS0.IWDTSTPCTL	IWDTCSTPR.SLCSTP

28.4 Link Operation by the Event Link Function

The IWDT is capable of link operation for a specified module when an interrupt request signal is used as an event signal. The IWDT outputs an event signal on an underflow of the down-counter or generation of a refresh error.

It outputs an event signal regardless of the setting of the reset interrupt request select bit (IWDTRCR.RSTIRQS) in register start mode or the IWDT reset interrupt request select bit (OFS0.IWDTRSTIRQS) in auto start mode. It can output an event signal at generation of the next interrupt source while retaining the refresh error flag (IWDTSR.REFEF) or underflow flag (IWDTSR.UNDFF).

28.5 Usage Notes

28.5.1 Transitioning to Low Power Mode

The refresh procedure must be completed before transitioning to low power mode. For details, see section 28.3.3. Refresh Operation.

28.5.2 Restriction on reading the counter value after returning from Deep Software Standby mode 1

After returning from Deep Software Standby mode 1, the counter value may not be read correctly and may read 0. To read the correct counter value after returning from Deep Software Standby mode 1, one of the following is required:

- Discard the read value until a value other than 0 is read.
- Wait one countdown period before reading the counter value.
- Refresh the counter when the refresh-permitted period is set to 100%.